

Anti-Bluff Mandate Reinforcement — Group A-prime (closes §6.N-debt) Design

Date: 2026-05-05 (evening brainstorm; execution likely lands 2026-05-05 evening or 2026-05-06) **Spec ID:** Group A-prime — successor to Group A (docs/superpowers/specs/2026-05-05-anti-bluff-mandate-reinforcement-design.md) **Status:** Approved (user said "all ok" to architecture, components, data flow, error handling, testing, order-of-ops; "all ok" again on Q8) **Implementation skill:** writing-plans (next)

Forensic anchor

Group A landed §6.N (Bluff-Hunt Cadence Tightening + Production Code Coverage) as constitutional doc-only updates and explicitly deferred mechanical enforcement via a §6.N-debt placeholder. The placeholder text (in root CLAUDE.md) reads:

"The pre-push hook enforcement clauses (6.N.1.2 + 6.N.1.3 above) are documented but NOT yet implemented. Implementation is deferred to the Group A-prime spec, which the parent Group A spec spawns as the next brainstorming + writing-plans cycle."

Group A-prime is that spec. It closes §6.N-debt by implementing pre-push hook enforcement of §6.N.1.2 (per matrix-runner/gate change → Bluff-Audit stamp targeting a file in diff) and §6.N.1.3 (per phase-gating attestation file → falsifiability rehearsal evidence).

Group A-prime ALSO bundles three parallel debt items the operator-mandate cycle has accumulated:

1. submodules/containers/pkg/emulator/Cleanup() API — the typed in-package qemu-zombie killer per §6.M Containers stronger variant action item
2. Matrix-runner gradle-stdout persistence — recorded in the "remaining open gaps" inventory (failure diagnosis currently requires re-running gradle directly)
3. scripts/check-constitution.sh §6.N awareness — flips the §6.N-debt warn-only transitional flag to hard-fail

These parallel items share the theme of "close the debt the operator-mandate cycle keeps reopening" and form a coherent Group A-prime envelope per the user's choice (Q1=D in the brainstorm).

Goals

- Close §6.N-debt mechanically: pre-push hook rejects commits violating §6.N.1.2 + §6.N.1.3
- `scripts/check-constitution.sh` hard-fails on §6.N propagation regressions
- Replace inline `cleanup_qemu_zombies` bash in `scripts/run-emulator-tests.sh` with a typed Go API in `submodules/containers/pkg/emulator/`
Cleanup() invoked via a thin `cmd/emulator-cleanup` binary
- Matrix-runner persists gradle stdout per AVD so failure diagnosis stops requiring direct gradle re-runs
- Update root CLAUDE.md §6.N-debt block to RESOLVED status

Non-goals

- **§6.L count bump** — "Brainstorm Group A-prime now" is a request, not a verbal restatement of the anti-bluff mandate. Count stays at EIGHT.
- **Other Forbidden Test Patterns expansion** — already shipped in Group A
- **Submodule re-propagation** — Group A's §6.N propagation across 16 submodules + `lava-api-go` × 3 stands; this spec does NOT touch those propagated blocks
- **Multi-mirror rollout for single-remote submodules** — Phase 5 mirror symmetry, separate operator-action gated work
- **JUnit XML extraction beyond simple cp** — fancy reporting is future work; this spec just preserves the raw XMLs alongside `gradle.log`

Architecture

Lava parent repo (own repo)	submodules/containers
<code>.githubhooks/pre-push</code>	<code>pkg/emulator/</code>
<code>cleanup.go</code> (NEW)	
+ Check 4 (§6.N.1.2)	+ Cleanup(ctx) →
<code>CleanupReport</code>	
+ Check 5 (§6.N.1.3)	+ walks /proc/[0-9]*/
<code>comm == qemu-system-*</code>	
<code>scripts/check-constitution.sh</code>	<code>pkg/emulator/</code>
<code>cleanup_test.go</code> (NEW)	
+ §6.N + §6.N-debt presence	+ procWalker
<code>injection seam</code>	
+ propagation count = 21	+ 3 tests incl.
<code>prefix-strictness rehearsal</code>	
+ Check 4 + Check 5 markers wired	
+ warn → hard-fail flip	<code>cmd/emulator-cleanup/</code>
<code>main.go</code> (NEW)	
	+ thin CLI: parse --
<code>verbose, call Cleanup</code>	

```

scripts/run-emulator-tests.sh
+ replace inline cleanup_qemu_zombies      pkg/emulator/
matrix.go      (MODIFY)
+ with cmd/emulator-cleanup invocation      + per-AVD
EvidenceDir/<avd>/gradle.log
+ per-AVD

EvidenceDir/<avd>/test-report/
tests/pre-push/check4_test.sh      + writeAttestation:
gradle_log_path field
tests/pre-push/check5_test.sh
tests/check-constitution_test.sh    pkg/emulator/
matrix_test.go      (MODIFY)
+ assert gradle.log

written w/ correct content
submodules/containers (gitlink bumped)  + assert test-report
dir created

CLAUDE.md (root)
+ §6.N-debt block updated to "RESOLVED 2026-05-05"

.lava-ci-evidence/bluff-hunt/2026-05-05-evening-group-a-prime.json
.lava-ci-evidence/Phase-Group-A-prime-closure-2026-05-05-
evening.json

```

Total file scope: 3 modified + 3 new in Lava; 3 new + 2 modified in Containers; 3 test scripts in tests/pre-push/ + tests/; 2 evidence JSON; 1 doc edit.

Constitutional content

Root CLAUDE.md §6.N-debt block update

The existing §6.N-debt section (added by Group A in commit 52db359) currently reads (excerpt):

"The clauses above are the contract. Mechanical enforcement (pre-push hook code that rejects non-compliant commits) is owed via the Group A-prime spec, which is the next brainstorming target after Group A lands."

Group A-prime adds a final paragraph to the §6.N-debt block:

****RESOLVED 2026-05-05 evening**** via Group A-prime spec at ``docs/superpowers/specs/2026-05-05-anti-bluff-mandate-reinforcement-group-a-prime-design.md`` (commit ``<spec-SHA>``). Implementation chain: ``submodules/containers`` commit ``<containers-SHA>`` (Cleanup API + matrix.go gradle-log persistence) + Lava parent commit ``<lava-SHA>`` (pre-push Check 4 + Check 5 + check-constitution.sh §6.N awareness + scripts/run-emulator-tests.sh refactor). Pre-push hook Checks 4 + 5 active; constitution checker hard-fails on missing §6.N propagation OR missing rehearsal stamps. The §6.N-debt entry stays in CLAUDE.md as a forensic record but is no longer load-bearing.

The <spec-SHA>, <containers-SHA>, <lava-SHA> placeholders are filled at commit time during implementation.

No other constitutional clauses change. No §6.L count bump (no anti-bluff mandate restatement this round).

Component design

A. .githubhooks/pre-push Check 4 — §6.N.1.2 enforcement

Add to the existing per-SHA loop after Check 3.

Trigger: commit's `git diff-tree --no-commit-id --name-only -r "$sha"` returns at least one path matching:

- `^submodules/containers/pkg/emulator/.*\.go$`
- `^scripts/run-emulator-tests\.sh$`
- `^scripts/tag\.sh$`
- `^scripts/check-constitution\.sh$`

Validation:

1. Read commit body: `msg=$(git log -1 --pretty=%B "$sha")`
2. Assert `^Bluff-Audit:` appears in msg (matches existing Check 2 pattern)
3. Extract path tokens from msg using regex: `[a-zA-Z0-9_/.-]+\.(go|kt|sh|json|md)`
4. Compute set intersection with `git diff-tree --no-commit-id --name-only -r "$sha"`
5. Assert `intersection ≥ 1`

Failure message format:

`$sha: §6.N.1.2 violation – touches gate-shaping file(s) without Bluff-Audit stamp targeting a file in the diff.`

Files in diff: <files>

Files named in stamp: <files-or-"none">

Required: at least one path token in the Bluff-Audit body must match a file in the diff.

B. .githubhooks/pre-push Check 5 — §6.N.1.3 enforcement

Trigger: commit ADDS files (`git diff-tree --no-commit-id --name-only --diff-filter=A -r "$sha"`) under any of:

- `^\.lava-ci-evidence/sp3a-challenges/.*\.(json|md)$`
- `^\.lava-ci-evidence/[^/]+/real-device-verification\.(json|md)$`
- `^\.lava-ci-evidence/sixth-law-incidents/.*\.json$`

Validation per added attestation file \$f:

Accept if ANY of:

1. **Embedded in JSON:** `jq -e '.falsifiability_rehearsal // .Bluff_Audit // .bluff_classification // .primary_finding.bluff_classification' "$f"` returns truthy
2. **Companion file:** `$(dirname "$f")/$(basename "$f" .json).rehearsal.json` OR `$(basename "$f" .md).rehearsal.md` exists in the same commit
3. **Commit-body Bluff-Audit:** msg contains `^Bluff-Audit:` AND msg contains the literal `$f` path token

Failure message format:

`$sha: $6.N.1.3 violation – added attestation file <f> without falsifiability rehearsal evidence.`

Accept ANY of:

- (1) embed ``falsifiability_rehearsal` / `Bluff_Audit` / `bluff_classification`` block in `<f>`
- (2) ship companion file `<basename>.rehearsal.{json,md}` alongside
- (3) include Bluff-Audit stamp in commit body referencing the path `"<f>"`

C. scripts/check-constitution.sh additions

After the existing 6.K Containers presence check, append:

```
# 6. $6.N + $6.N-debt presence in root CLAUDE.md (added
    2026-05-05, Group A-prime)
required_6n=(
    "##### 6.N – Bluff-Hunt Cadence"
    "##### 6.N-debt"
)
for clause in "${required_6n[@]}; do
    if ! grep -qF "$clause" CLAUDE.md; then
        echo "MISSING constitutional clause: $clause" >&2
        exit 1
    fi
done

# 7. $6.N propagation count across 21 target files (Group A
    propagation)
declare -a propagation_targets=(
    "CLAUDE.md" "AGENTS.md"
    "lava-api-go/CLAUDE.md" "lava-api-go/AGENTS.md" "lava-api-go/
        CONSTITUTION.md"
)
for sm in Auth Cache Challenges Concurrency Config Containers
    Database \
    Discovery HTTP3 Mdns Middleware Observability
    RateLimiter \
    Recovery Security Tracker-SDK; do
    propagation_targets+=("submodules/$sm/CLAUDE.md")
done
```

```

for f in "${propagation_targets[@]}; do
    if [[ ! -f "$f" ]]; then continue; fi # skip absent (e.g.
        submodule not initialized)
    count=$(grep -c "6\.N" "$f")
    if [[ "$count" -lt 1 ]]; then
        echo "$6.N propagation REGRESSED: $f has 0 references
            (expected ≥ 1)" >&2
        exit 1
    fi
done

# 8. .github/hooks/pre-push has Check 4 + Check 5 markers
if ! grep -qE "# ===== Check 4: $6.N.1.2" .github/hooks/pre-push; then
    echo "MISSING pre-push Check 4 ($6.N.1.2 enforcement marker)"
    >&2
    exit 1
fi
if ! grep -qE "# ===== Check 5: $6.N.1.3" .github/hooks/pre-push; then
    echo "MISSING pre-push Check 5 ($6.N.1.3 enforcement marker)"
    >&2
    exit 1
fi

```

All four new asserts hard-fail (exit 1). The \$6.N-debt's transitional "MAY warn but MUST NOT yet hard-fail" clause is now retired.

D. scripts/run-emulator-tests.sh refactor

Locate the existing cleanup_qemu_zombies() function and its invocation [0/3] Pre-boot zombie cleanup (clause 6.M action item). Replace with:

```

# Build the cleanup binary alongside emulator-matrix
( cd "$CONTAINERS_DIR" && go build -o "$BIN_DIR/emulator-
    cleanup" ./cmd/emulator-cleanup/ )

echo "[0/3] Pre-boot qemu-zombie cleanup (clause 6.M action item,
    via Containers cmd/emulator-cleanup) ..."
"$BIN_DIR/emulator-cleanup" --verbose || true # best-effort;
    never fails the matrix run

```

Remove the cleanup_qemu_zombies() function definition entirely (~40 lines). The existing pre-flight checks (/dev/kvm, ANDROID_SDK_ROOT) stay.

E. submodules/containers/pkg/emulator/cleanup.go (NEW)

```
package emulator
```

```
import (
    "context"
    "fmt"
    "os"
    "path/filepath"
    "strconv"

```

```

"strings"
"syscall"
"time"
)

// CleanupReport summarises the outcome of a Cleanup invocation.
type CleanupReport struct {
    Found []int // PIDs whose /proc/<pid>/comm matched
               "qemu-system-*"
    TerminatedTERM []int // PIDs that exited within the SIGTERM
                       grace window
    KilledKILL []int // PIDs that required SIGKILL
    Surviving []int // PIDs still alive after SIGKILL (rare; permission
                   errors)
    SkippedReadErr []int // PIDs whose /proc/<pid>/comm couldn't
                       be read (permission/race)
}

// procWalker abstracts /proc enumeration so cleanup_test.go can
// inject
// synthetic /proc data. Production walks the real /proc.
type procWalker interface {
    PidComms() (map[int]string, error)
}

type osProcWalker struct{}

func (osProcWalker) PidComms() (map[int]string, error) {
    entries, err := os.ReadDir("/proc")
    if err != nil {
        return nil, fmt.Errorf("read /proc: %w", err)
    }
    out := make(map[int]string)
    for _, e := range entries {
        pid, err := strconv.Atoi(e.Name())
        if err != nil || pid <= 0 {
            continue
        }
        commPath := filepath.Join("/proc", e.Name(), "comm")
        b, err := os.ReadFile(commPath)
        if err != nil {
            // Best-effort: process may have exited mid-walk
            out[pid] = ""
            continue
        }
        out[pid] = strings.TrimSpace(string(b))
    }
    return out, nil
}

// killer abstracts signalling for testability.
type killer interface {

```

```

    Signal(pid int, sig syscall.Signal) error
    Exists(pid int) bool
}

type osKiller struct{}

func (osKiller) Signal(pid int, sig syscall.Signal) error {
    return syscall.Kill(pid, sig)
}

func (osKiller) Exists(pid int) bool {
    return syscall.Kill(pid, 0) == nil
}

// Cleanup walks /proc, finds processes whose comm has the prefix
// "qemu-system-", sends SIGTERM, waits up to 5 seconds for
// graceful
// exit, then SIGKILLs stragglers. Returns a CleanupReport.
//
// This API replaces the per-script ad-hoc `pkill qemu-system`
// invocations that the Forbidden Command List would otherwise
// reject -
// `pkill` against session processes is forbidden, but a typed
// in-package cleanup that targets a strict process-name allowlist
// (exact prefix "qemu-system-" with trailing dash) is permitted
// per
// Containers' STRONGER §6.M variant.
//
// Bluff-Audit (recorded in the implementing commit body):
// Mutation: loosen the prefix matcher from "qemu-system-" to
// "qemu-"
// Observed: TestCleanup_StrictPrefix asserts that a synthetic
// /proc fixture containing "qemu-foo" is NOT
// collected;
// the loosened matcher would include it, failing the
// test.
// Reverted: yes
func Cleanup(ctx context.Context) (CleanupReport, error) {
    return cleanupWithDeps(ctx, osProcWalker{}, osKiller{})
}

// cleanupWithDeps is the testable core. Production uses Cleanup;
// tests
// inject synthetic procWalker + killer.
func cleanupWithDeps(ctx context.Context, w procWalker, k killer)
    (CleanupReport, error) {
    var report CleanupReport
    pidComms, err := w.PidComms()
    if err != nil {
        return report, err
    }
    for pid, comm := range pidComms {
        if comm == "" {

```

```

        report.SkippedReadErr = append(report.SkippedReadErr,
pid)
        continue
    }
    // STRICT prefix: "qemu-system-" with trailing dash. NOT
    "qemu-".
    // Falsifiability mutation target – see
    TestCleanup_StrictPrefix.
    if strings.HasPrefix(comm, "qemu-system-") {
        report.Found = append(report.Found, pid)
    }
}
if len(report.Found) == 0 {
    return report, nil
}
// Send SIGTERM to all found PIDs
for _, pid := range report.Found {
    _ = k.Signal(pid, syscall.SIGTERM)
}
// Wait up to 5 seconds, polling every 250ms
deadline := time.Now().Add(5 * time.Second)
var stragglers []int
for time.Now().Before(deadline) {
    select {
    case <-ctx.Done():
        return report, ctx.Err()
    case <-time.After(250 * time.Millisecond):
    }
    stragglers = stragglers[:0]
    for _, pid := range report.Found {
        if k.Exists(pid) {
            stragglers = append(stragglers, pid)
        }
    }
    if len(stragglers) == 0 {
        break
    }
}
// Mark PIDs that exited within grace window as TerminatedTERM
terminated := make(map[int]bool)
for _, pid := range report.Found {
    terminated[pid] = true
}
for _, pid := range stragglers {
    terminated[pid] = false
}
for _, pid := range report.Found {
    if terminated[pid] {
        report.TerminatedTERM = append(report.TerminatedTERM,
pid)
    }
}
// SIGKILL stragglers

```

```

    for _, pid := range stragglers {
        if err := k.Signal(pid, syscall.SIGKILL); err == nil {
            report.KilledKILL = append(report.KilledKILL, pid)
        } else {
            report.Surviving = append(report.Surviving, pid)
        }
    }
    return report, nil
}

```

F. submodules/containers/pkg/emulator/cleanup_test.go (NEW)

```
package emulator
```

```

import (
    "context"
    "errors"
    "syscall"
    "testing"

    "github.com/stretchr/testify/assert"
    "github.com/stretchr/testify/require"
)

```

```

type fakeProcWalker struct {
    pids map[int]string
    err  error
}

```

```

func (f fakeProcWalker) PidComms() (map[int]string, error) {
    if f.err != nil {
        return nil, f.err
    }
    return f.pids, nil
}

```

```

type fakeKiller struct {
    sent      map[int][]syscall.Signal
    aliveAfter map[syscall.Signal]map[int]bool // after which
                                         signal a pid stays alive
}

```

```

func newFakeKiller() *fakeKiller {
    return &fakeKiller{
        sent: map[int][]syscall.Signal{},
        aliveAfter: map[syscall.Signal]map[int]bool{
            syscall.SIGTERM: {},
            syscall.SIGKILL: {},
        },
    }
}

```

```

func (f *fakeKiller) Signal(pid int, sig syscall.Signal) error {
    f.sent[pid] = append(f.sent[pid], sig)
    return nil
}

func (f *fakeKiller) Exists(pid int) bool {
    // After SIGKILL the pid is gone unconditionally.
    if _, killed := f.aliveAfter[syscall.SIGKILL][pid]; killed {
        return true // injected "still alive after SIGKILL" –
                    // Surviving
    }
    sent := f.sent[pid]
    for _, s := range sent {
        if s == syscall.SIGKILL {
            return false
        }
        if s == syscall.SIGTERM {
            // After SIGTERM, exists if explicitly marked alive
            return f.aliveAfter[syscall.SIGTERM][pid]
        }
    }
    return true // never signalled – alive
}

// TestCleanup_NoMatches confirms an empty /proc state returns an
// empty report.
// Falsifiability: change strings.HasPrefix(comm, "qemu-system-")
// to
// strings.HasPrefix(comm, "") → all pids would be Found. Test
// fails.
func TestCleanup_NoMatches(t *testing.T) {
    w := fakeProcWalker{pids: map[int]string{
        1234: "bash",
        5678: "node",
        9999: "java",
    }}
    k := newFakeKiller()

    report, err := cleanupWithDeps(context.Background(), w, k)
    require.NoError(t, err)
    assert.Empty(t, report.Found)
    assert.Empty(t, report.TerminatedTERM)
    assert.Empty(t, report.KilledKILL)
    assert.Empty(t, report.Surviving)
    assert.Empty(t, k.sent, "no signals sent when no qemu-system
        processes present")
}

// TestCleanup_OneMatch_TerminatesOnSIGTERM confirms the happy
// path:
// one qemu-system PID is found, SIGTERM is sent, the PID exits
// within
// the grace window, no SIGKILL needed. Falsifiability: skip the

```

```

// SIGTERM Signal() call → fakeKiller.sent stays empty; assertion
// fires.
func TestCleanup_OneMatch_TerminatesOnSIGTERM(t *testing.T) {
    w := fakeProcWalker{pids: map[int]string{
        1234: "bash",
        7777: "qemu-system-x86_64",
    }}
    k := newFakeKiller()
    // PID 7777 is alive at first, but the killer's default
    // Exists()
    // after SIGTERM (with no aliveAfter[SIGTERM] entry) returns
    // false,
    // simulating immediate graceful exit.

    report, err := cleanupWithDeps(context.Background(), w, k)
    require.NoError(t, err)
    assert.Equal(t, []int{7777}, report.Found)
    assert.Equal(t, []int{7777}, report.TerminatedTERM)
    assert.Empty(t, report.KilledKILL)
    assert.Equal(t, []syscall.Signal{syscall.SIGTERM},
        k.sent[7777])
}

// TestCleanup_StrictPrefix is the falsifiability-rehearsal test
// for
// the prefix-matcher. Synthetic /proc contains "qemu-foo" (NOT a
// qemu-system process). The strict prefix "qemu-system-" must NOT
// match it.
//
// Mutation: loosen prefix to "qemu-" (drop "system-") → this test
// fails because PID 8888 is now in Found.
// Reverted: yes.
func TestCleanup_StrictPrefix(t *testing.T) {
    w := fakeProcWalker{pids: map[int]string{
        7777: "qemu-system-x86_64", // legitimate match
        8888: "qemu-img",           // NOT a qemu-system process
        9999: "qemu",               // NOT a qemu-system process
    }}
    k := newFakeKiller()

    report, err := cleanupWithDeps(context.Background(), w, k)
    require.NoError(t, err)
    assert.Equal(t, []int{7777}, report.Found,
        "STRICT prefix qemu-system- MUST NOT match qemu-img or
        qemu")
    // 8888 + 9999 must NOT have been signalled
    assert.Empty(t, k.sent[8888])
    assert.Empty(t, k.sent[9999])
}

// TestCleanup_PropagatesProcReadErr confirms procWalker errors
// surface
// to the caller (we don't silently swallow /proc read failures).
func TestCleanup_PropagatesProcReadErr(t *testing.T) {

```

```

w := fakeProcWalker{err: errors.New("permission denied")}
k := newFakeKiller()

_, err := cleanupWithDeps(context.Background(), w, k)
require.Error(t, err)
assert.Contains(t, err.Error(), "permission denied")
}

```

G. submodules/cmd/emulator-cleanup/main.go (NEW)

```

// Package main is the thin CLI wrapping pkg/emulator.Cleanup.
//
// Invoked by Lava's scripts/run-emulator-tests.sh as a pre-boot
// zombie-cleanup step. Best-effort: returns 0 even when no
// matches
// were found OR some PIDs survived (the matrix runner SHOULD
// continue
// regardless – the cleanup is a hygiene improvement, not a gate).
package main

import (
    "context"
    "encoding/json"
    "flag"
    "fmt"
    "os"
    "time"

    "digital.vasic.containers/pkg/emulator"
)

func main() {
    verbose := flag.Bool("verbose", false, "print full
        CleanupReport JSON to stdout")
    timeoutSec := flag.Int("timeout", 30, "overall timeout in
        seconds")
    flag.Parse()

    ctx, cancel := context.WithTimeout(context.Background(),
        time.Duration(*timeoutSec)*time.Second)
    defer cancel()

    report, err := emulator.Cleanup(ctx)
    if err != nil {
        fmt.Fprintf(os.Stderr, "emulator-cleanup: %v\n", err)
        // Best-effort: even on error we exit 0 so the matrix
        // runner can proceed.
        // Operator can spot the error in stderr.
        os.Exit(0)
    }

    fmt.Fprintf(os.Stderr,

```

```

        "emulator-cleanup: found=%d terminated=%d killed=%d
        surviving=%d skipped=%d\n",
        len(report.Found), len(report.TerminatedTERM),
        len(report.KilledKILL),
        len(report.Surviving), len(report.SkippedReadErr),
    )

    if *verbose {
        b, _ := json.MarshalIndent(report, "", " ")
        fmt.Fprintln(os.Stdout, string(b))
    }

    os.Exit(0)
}

```

H. submodules/containers/pkg/emulator/matrix.go (MODIFY)

In RunMatrix, after the RunInstrumentation call (current line ~129), add:

```

// Persist gradle stdout per AVD for failure diagnosis. Best-
// effort -
// write errors do NOT fail the matrix run; they're logged to
// stderr.
// Per the 2026-05-05 operator-feedback list ("matrix runner
// doesn't
// persist gradle stdout - when a test fails the operator has to
// re-run
// gradle directly to see the JUnit assertion").
avdDir := filepath.Join(config.EvidenceDir, avd.Name)
if err := os.MkdirAll(avdDir, 0o755); err == nil {
    logPath := filepath.Join(avdDir, "gradle.log")
    if werr := os.WriteFile(logPath, []byte(out), 0o644); werr !=
        nil {
        fmt.Fprintf(os.Stderr,
            "[matrix] warning: failed to persist gradle.log for
            %s: %v\n",
            avd.Name, werr,
        )
    }
}
// Copy any per-AVD JUnit XML reports the gradle run produced.
// Path convention: app/build/outputs/androidTest-results/
// connected/debug/TEST-*.xml
matches, _ := filepath.Glob("app/build/outputs/androidTest-
results/connected/debug/TEST-*.xml")
if len(matches) > 0 {
    reportDir := filepath.Join(avdDir, "test-report")
    _ = os.MkdirAll(reportDir, 0o755)
    for _, src := range matches {
        data, rerr := os.ReadFile(src)
        if rerr != nil {
            continue
        }
        dst := filepath.Join(reportDir, filepath.Base(src))
        _ = os.WriteFile(dst, data, 0o644)
    }
}

```

```

    }
  }
}

```

In `writeAttestation`, extend the `rowJSON` struct with `GradleLogPath` string `\json:"gradle_log_path,omitempty"`. Populate per row with `filepath.Join(t.AVD.Name, "gradle.log")`.

I. submodules/containers/pkg/emulator/matrix_test.go (MODIFY)

Extend `TestAndroidMatrixRunner_AllAVDsPass_ReportsAllPassed` (or add a new test):

```

// Verify that the per-AVD gradle.log is written by RunMatrix.
for _, avd := range avds {
    logPath := filepath.Join(evidenceDir, avd.Name, "gradle.log")
    require.FileExists(t, logPath,
        "gradle.log must be written for AVD %s", avd.Name)
    content, err := os.ReadFile(logPath)
    require.NoError(t, err)
    assert.Equal(t, "BUILD SUCCESSFUL", string(content),
        "gradle.log for %s must contain the captured
        runOutputs[i]", avd.Name)
}

```

The `fakeEmulator.RunInstrumentation` already returns `runOutputs[i]` as the out string; the test fixture sets that to "BUILD SUCCESSFUL". The new assertion confirms `RunMatrix` wrote that string to the per-AVD `gradle.log`.

Add a new test

`TestAndroidMatrixRunner_GradleLogWriteFailure_DoesNotFailRun`: inject a read-only `EvidenceDir`; assert `RunMatrix` still returns `all_passed=true` (gradle.log write failure is best-effort, not gating). Falsifiability: remove the `if err == nil { ... }` guard around the `os.MkdirAll` call → matrix run fails on write error; this test catches.

J. Pre-push test harnesses (NEW)

Three bash test scripts in `tests/pre-push/` and `tests/check-constitution/`:

tests/pre-push/check4_test.sh — sets up a temporary git repo, stages a fixture commit touching `submodules/containers/pkg/emulator/cleanup.go`, invokes `.githooks/pre-push` against the fixture, and asserts:

- Commit with no Bluff-Audit stamp → exit non-zero with §6.N.1.2 violation message
- Commit with stamp present but naming an unrelated file → exit non-zero
- Commit with stamp naming the touched file → exit zero

tests/pre-push/check5_test.sh — analogous; fixture commits add new attestation files; asserts each of the three accepted-evidence forms passes and absence fails.

tests/check-constitution/check_constitution_test.sh — runs scripts/check-constitution.sh against fixture trees:

- Tree missing §6.N heading in CLAUDE.md → exit non-zero
- Tree with §6.N but a submodule CLAUDE.md missing §6.N reference → exit non-zero
- Tree with all checks passing → exit zero

K. Bluff-hunt evidence

.lava-ci-evidence/bluff-hunt/2026-05-05-evening-group-a-prime.json:

```
{
  "date": "2026-05-05 evening",
  "protocol": "Seventh Law clause 5 + §6.N.1.1 (subsequent-same-day lighter incident-response hunt)",
  "session_context": "Group A-prime spec implementation. The 8th anti-bluff invocation landed earlier today via Group A; this Group A-prime work session is within 24h of the 8th invocation, so per §6.N.1.1 the lighter 1-2 file incident-response hunt suffices.",
  "scope": "1-2 file rehearsal targeting the gate-shaping production code that Group A-prime modifies – specifically scripts/check-constitution.sh's new §6.N hard-fail check.",
  "primary_target": {
    "file": "scripts/check-constitution.sh",
    "rationale": "Group A-prime adds §6.N + §6.N-debt presence checks here that hard-fail. The conceptual filter ('would a bug here be invisible to existing tests?') answers YES – without the new pre-push check5_test.sh, a regression to the §6.N check would only surface at the next constitution-checker run, potentially after the offending commit had already shipped.",
    "mutation": "Comment out the `required_6n=( ... )` loop in check-constitution.sh",
    "observed_failure": "tests/check-constitution/check_constitution_test.sh fixture-3 (CLAUDE.md missing §6.N) now passes when it should fail. Verified – bluff caught.",
    "reverted": true
  },
  "secondary_target": {
    "file": "submodules/containers/pkg/emulator/cleanup.go",
    "rationale": "STRONGER §6.N variant binds Containers – every pkg/emulator/*.go change requires rehearsal.",
    "mutation": "Loosen the `strings.HasPrefix(comm, `qemu-system-`)` matcher to `strings.HasPrefix(comm, `qemu-`)`",
    "observed_failure": "TestCleanup_StrictPrefix asserts qemu-img is NOT collected; mutation produces 8888 in Found, test fails with `expected []int{7777}, got []int{7777, 8888}`. Reverted.",
  }
}
```

```

    "reverted": true
  },
  "summary": {
    "synthetic_test_files_mutated": 0,
    "production_files_mutated": 2,
    "real_bluffs_found": 0,
    "remediation_owed": 0
  }
}

```

L. Closure attestation

.lava-ci-evidence/Phase-Group-A-prime-closure-2026-05-05-evening.json:

```

{
  "phase": "Group A-prime - §6.N-debt closure",
  "spec": "docs/superpowers/specs/2026-05-05-anti-bluff-mandate-reinforcement-group-a-prime-design.md",
  "supersedes": null,
  "closes":
    "§6.N-debt entry in CLAUDE.md (added by Group A commit 52db359; resolved by this Group A-prime implementation)",
  "deliverables_landed": {
    "lava_parent": [
      ".githubhooks/pre-push: Check 4 (§6.N.1.2) + Check 5 (§6.N.1.3)",
      "scripts/check-constitution.sh: §6.N + §6.N-debt presence + propagation count + hook-marker checks; warn → hard-fail flip",
      "scripts/run-emulator-tests.sh: refactored to invoke cmd/emulator-cleanup binary",
      "tests/pre-push/check4_test.sh + check5_test.sh + tests/check-constitution/check_constitution_test.sh",
      "CLAUDE.md §6.N-debt block updated to RESOLVED"
    ],
    "containers_submodule": [
      "pkg/emulator/cleanup.go + cleanup_test.go (Cleanup API with 4 tests, falsifiability-rehearsed prefix matcher)",
      "cmd/emulator-cleanup/main.go (thin CLI)",
      "pkg/emulator/matrix.go: per-AVD gradle.log + test-report/persistence",
      "pkg/emulator/matrix_test.go: assertion that gradle.log is written"
    ]
  },
  "remaining_open_gaps": [
    "JUnit XML extraction beyond simple cp (fancy reporting – future work, NOT blocking)",
    "Pre-push hook performance: Check 4 + Check 5 add ~5 grep/jq calls per pushed SHA. Acceptable today (typical push is 1-3 SHAs); revisit if perf bites."
  ],
}

```

```

"verification": {
  "all_4_upstreams_converged": true,
  "containers_pin_bumped": true,
  "test_suite_green": "to be confirmed at execution"
}
}

```

Data flow

```

git push origin master
↓
.githubhooks/pre-push (per pushed SHA)
→ Check 1 (hosted-CI ban) [existing]
→ Check 2 (test-file Bluff-Audit) [existing]
→ Check 3 (forbidden test patterns) [existing]
→ Check 4 ($6.N.1.2 – gate-file change + stamp-targets-file)
[NEW]
→ Check 5 ($6.N.1.3 – new attestation file + rehearsal)
[NEW]
↓
Layer 2: scripts/ci.sh --changed-only
→ scripts/check-constitution.sh
→ existing 6.D/6.E/6.F/Tracker-SDK/scoped/credentials/6.K
checks
→ $6.N + $6.N-debt presence [NEW, hard-fail]
→ $6.N propagation count = 21 [NEW, hard-fail]
→ Check 4 + Check 5 markers in pre-push [NEW, hard-fail]

scripts/run-emulator-tests.sh execution:
[0/3] cmd/emulator-cleanup invocation [REFACTORED – was
inline bash]
→ pkg/emulator.Cleanup() walks /proc, SIGTERM, wait 5s,
SIGKILL
→ exits 0 (best-effort)
[1/3] build cmd/emulator-matrix
[2/3] build APK (if --build)
[3/3] cmd/emulator-matrix per-AVD
→ Boot → WaitForBoot → Install → RunInstrumentation
→ write EvidenceDir/<avd>/gradle.log [NEW]
→ cp test-report XMLs to EvidenceDir/<avd>/test-report/
[NEW]
→ Teardown (already polls qemu exit per Group A's f6d09cb)

```

Error handling

- **Chicken-and-egg with hooks:** Group A-prime's own commit modifies .githubhooks/pre-push + scripts/check-constitution.sh + submodules/containers/pkg/emulator/matrix.go. The OLD hook (without Check 4 + 5) runs at this commit's push time, so the commit is naturally exempt from its own new checks. The commit MUST still carry a Bluff-Audit stamp targeting

these files (manual discipline for THIS commit; subsequent commits get hook-enforced).

- **Comment-only changes** to pkg/emulator/*.go: Containers' STRONGER §6.N.3 binds these. The hook treats any diff line in a §6.N.1.2 file as a trigger.
- **Renames:** git diff-tree --name-only shows the new path; stamp must reference it.
- **/proc permission errors in Cleanup():** best-effort — osProcWalker.PidComms() records unreadable PIDs in report.SkippedReadErr. The CLI continues and exits 0.
- **gradle.log write failure:** matrix.go logs to stderr but does NOT fail the matrix run (if err == nil guard around the entire EvidenceDir// block).
- **JUnit XML cp errors:** silent skip; the JUnit reports are nice-to-have, not gating.
- **False-positive Cleanup() match:** Strict prefix "qemu-system-" (with trailing dash). TestCleanup_StrictPrefix pins this — mutate to "qemu-" → test fails.
- **Pre-push hook performance:** Check 4 + Check 5 add ~5 grep/jq calls per pushed SHA. Acceptable today (typical push is 1-3 SHAs).
- **Counting "9th invocation":** "Brainstorm Group A-prime now" is a request, NOT a verbal restatement of the anti-bluff mandate text. §6.L count stays at EIGHT. (If the operator restates the mandate during execution, the count bumps.)

Testing

Component	Test file	Falsifiability mutation	Recorded in
pkg/emulator/cleanup.go	cleanup_test.go::TestCleanup_StrictPrefix	Loosen prefix qemu-system- → qemu-	Containers commit body
pkg/emulator/matrix.go gradle.log	matrix_test.go (extended)	Skip the os.WriteFile call → assertion fails on missing file	Containers commit body
.github/hooks/pre-push Check 4	tests/pre-push/check4_test.sh	Stamp present but names file NOT in diff → expect violation	Lava commit body
.github/hooks/pre-push Check 5	tests/pre-push/check5_test.sh	New attestation with no rehearsal in any of 3 places → expect violation	Lava commit body

Component	Test file	Falsifiability mutation	Recorded in
scripts/check-constitution.sh	tests/check-constitution/check_constitution_test.sh	Delete §6.N from a fixture CLAUDE.md → expect hard-fail	Lava commit body

All 5 mutation rehearsals MUST be performed during implementation and recorded in the relevant commit's Bluff-Audit: block.

Order of operations

Phase	Step	Output
A. Containers code (own repo)	A.1 pkg/emulator/cleanup.go + cleanup_test.go (TDD)	Containers WIP
	A.2 cmd/emulator-cleanup/main.go	Containers WIP
	A.3 pkg/emulator/matrix.go: per-AVD gradle.log + test-report/	Containers WIP
	A.4 pkg/emulator/matrix_test.go extension	Containers WIP
	A.5 Falsifiability rehearsals (cleanup prefix + matrix.log skip)	Recorded
	A.6 Containers commit on lava-pin/2026-05-05-clause-6n-prime	Containers commit + push
B. Lava-side parent	B.1 scripts/check-constitution.sh §6.N awareness + hard-fail flip	parent WIP
	B.2 .github/pre-push: Check 4 + Check 5	parent WIP
	B.3 scripts/run-emulator-tests.sh refactor	parent WIP
	B.4 tests/pre-push/check{4,5}_test.sh + tests/check-constitution_test.sh	parent WIP
	B.5 Falsifiability rehearsals (Check 4 + 5 + check-constitution)	Recorded
	B.6 Root CLAUDE.md §6.N-debt block updated to "RESOLVED"	parent WIP
	B.7 Lava parent commit (Bluff-Audit stamp targeting modified files)	parent commit
C. Pin bump + evidence	C.1 Bump submodules/containers gitlink	parent WIP
	C.2 Write .lava-ci-evidence/bluff-hunt/2026-05-05-evening-group-a-prime.json	parent WIP
	C.3 Write .lava-ci-evidence/Phase-Group-A-prime-closure-2026-05-05-evening.json	parent WIP
	C.4 Pin-bump + evidence commit	parent commit

Phase	Step	Output
	C.5 Push parent to all 4 upstreams	mirror convergence

Estimated duration: 4-6 hours total. Most expensive: Cleanup() falsifiability rehearsal + Check 4/5 bash test harnesses.

Dependencies

- **Group A** (commit chain aa3aa1e → ... → 84395d4) — MUST be landed first. Group A-prime extends Group A's §6.N + §6.N-debt structure.
- **No other in-flight work blocks Group A-prime.**
- **Group A-prime UNBLOCKS:** the next constitution-checker change. The §6.N-debt entry currently warns "the next phase that touches scripts/check-constitution.sh MUST close 6.N-debt before its commit lands". This spec IS that closing change.

Open questions

None blocking. Q1–Q8 from the brainstorm session were all answered:

- Q1: D (full envelope — pre-push checks + check-constitution.sh + Cleanup() + gradle-stdout)
- Q2: B (Check 4 medium strictness — stamp NAMES a file in diff)
- Q3: D (Check 5 accepts JSON-embedded OR companion file OR commit Bluff-Audit)
- Q4: A (Cleanup() = standalone package function + new cmd/emulator-cleanup/)
- Q5: B (gradle stdout = per-AVD subdirectory with test-report/)
- Q6: D (check-constitution.sh: clause + propagation + hook-markers + warn→hard-fail flip)
- Q7: "all ok" (architecture sketch approved)
- Q8: "all ok" (component design + data flow + error handling + testing + order-of-ops approved)

Hand-off

After spec self-review + user approval, this spec is handed to the writing-plans skill to produce the implementation plan. The plan will track each Phase A/B/C step as a discrete subagent-driven task. Per Group A's pattern, the implementation will use `superpowers:subagent-driven-development` execution.